Estimation Theory

Linear Bayesian Estimators

Linear MMSE (LMMSE) or Linear Least Square Error (LLSE) Estimation

- Optimal Bayesian estimators are often too difficult to determine in closed form.
- They are too computationally intensive.
- Restrict estimators to be linear -> Linear MMSE estimator

LMMSE estimator is a linear estimator $$\hat{\theta} = \sum_{n=0}^{N-1} a_n x[n] + a_N$$

which minimizes the Bayesian MSE $$\mathrm{Bmse}(\hat{\theta}) = E\left[(\theta - \hat{\theta})^2\right]$$

Clearly, the LMMSE estimator is suboptimal unless the MMSE estimator is linear.

Example: LMMSE is not always feasible

$$x[0] \sim \mathcal{N}(0, \sigma^2) \qquad \text{Parameter: } \theta = x^2[0]. \text{ Estimate } \theta \text{ using } x[0].$$

$$\text{Perfect Estimator: } \hat{\theta} = x^2[0] \text{ (Bayesian MSE is zero).}$$

- Let us consider a linear estimator: $\hat{\theta} = a_0 x[0] + a_1$
- LMMSE estimator can be founded by minimizing

$$\begin{aligned} \mathrm{B}_{\mathrm{mse}}(\hat{\theta}) &= \mathbb{E}\left[(\theta - \hat{\theta})^2\right] \\ &= \mathbb{E}\left[(\theta - a_0 x[0] - a_1)^2\right] \end{aligned}$$

$$\frac{\partial}{\partial a_0}\mathrm{B}_{\mathrm{mse}}(\hat{\theta}) = 0 \implies \mathbb{E}\left[(\theta - a_0 x[0] - a_1)x[0]\right] = 0 \implies a_0 \mathbb{E}(x^2[0]) + \cancel{a_1 \mathbb{E}(x[0])} = \cancel{\mathbb{E}(\theta x[0])}$$

$$\frac{\partial}{\partial a_1}\mathrm{B}_{\mathrm{mse}}(\hat{\theta}) = 0 \implies \mathbb{E}\left[(\theta - a_0 x[0] - a_1)\right] = 0 \implies \cancel{a_0 \mathbb{E}(x[0])} + a_1 = \mathbb{E}(\theta)$$

$$\begin{aligned} a_0 &= 0 \\ a_1 &= \mathbb{E}(\theta) = \mathbb{E}(x^2[0]) = \sigma^2 \end{aligned} \qquad \implies \text{LMMSE Estimator: } \hat{\theta} = \sigma^2$$

$$\mathrm{B}_{\mathrm{mse}}(\hat{\theta}) = 2\sigma^4$$

No good!

<u>LMMSE Estimator</u>

$$\mathrm{Bmse}(\hat{\theta}) = E\left[(\theta - \hat{\theta})^2\right] \qquad \hat{\theta} = \sum_{n=0}^{N-1} a_n x[n] + a_N$$

$$\frac{\partial}{\partial a_N} E\left[\left(\theta - \sum_{n=0}^{N-1} a_n x[n] - a_N\right)^2\right] = -2E\left[\theta - \sum_{n=0}^{N-1} a_n x[n] - a_N\right]$$

$$\boxed{a_N = E(\theta) - \sum_{n=0}^{N-1} a_n E(x[n])}$$

$$\mathrm{Bmse}(\hat{\theta}) = E\left\{\left[\sum_{n=0}^{N-1} a_n (x[n] - E(x[n])) - (\theta - E(\theta))\right]^2\right\}$$

$$\begin{aligned}
\mathrm{Bmse}(\hat{\theta}) &= E\left\{\left[\mathbf{a}^T(\mathbf{x} - E(\mathbf{x})) - (\theta - E(\theta))\right]^2\right\} \\
&= E\left[\mathbf{a}^T(\mathbf{x} - E(\mathbf{x}))(\mathbf{x} - E(\mathbf{x}))^T\mathbf{a}\right] - E\left[\mathbf{a}^T(\mathbf{x} - E(\mathbf{x}))(\theta - E(\theta))\right] \\
&\quad - E\left[(\theta - E(\theta))(\mathbf{x} - E(\mathbf{x}))^T\mathbf{a}\right] + E\left[(\theta - E(\theta))^2\right] \\
&= \mathbf{a}^T\mathbf{C}_{xx}\mathbf{a} - \mathbf{a}^T\mathbf{C}_{x\theta} - \mathbf{C}_{\theta x}\mathbf{a} + C_{\theta\theta}
\end{aligned}$$

$$\frac{\partial \mathrm{Bmse}(\hat{\theta})}{\partial \mathbf{a}} = 2\mathbf{C}_{xx}\mathbf{a} - 2\mathbf{C}_{x\theta} \qquad \boxed{\mathbf{a} = \mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta}}$$

$$\begin{aligned}
\hat{\theta} &= \mathbf{a}^T\mathbf{x} + a_N \\
&= \mathbf{C}_{x\theta}^T\mathbf{C}_{xx}^{-1}\mathbf{x} + E(\theta) - \mathbf{C}_{x\theta}^T\mathbf{C}_{xx}^{-1}E(\mathbf{x})
\end{aligned}$$

LMMSE Estimator:

$$\boxed{\hat{\theta} = E(\theta) + \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x}))}$$

LMMSE Estimator

$$\mathrm{Bmse}(\hat{\theta}) = \mathbf{a}^T\mathbf{C}_{xx}\mathbf{a} - \mathbf{a}^T\mathbf{C}_{x\theta} - \mathbf{C}_{\theta x}\mathbf{a} + C_{\theta\theta} \qquad \mathbf{a} = \mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta}$$

$$\begin{aligned}\mathrm{Bmse}(\hat{\theta}) &= \mathbf{C}_{x\theta}^T\mathbf{C}_{xx}^{-1}\mathbf{C}_{xx}\mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta} - \mathbf{C}_{x\theta}^T\mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta} \\ &\quad - \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta} + C_{\theta\theta} \\ &= \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta} - 2\mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta} + C_{\theta\theta}\end{aligned}$$

$$\boxed{\mathrm{Bmse}(\hat{\theta}) = C_{\theta\theta} - \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta}}$$; Minimum Bayes MSE

LMMSE Estimator:

$$\boxed{\begin{aligned}&\hat{\theta} = E(\theta) + \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})) \\ &\mathrm{Bmse}(\hat{\theta}) = C_{\theta\theta} - \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta}\end{aligned}}$$

cf. MMSE estimator for a multivariate Gaussian r.v.

When $\mathbb{E}(\theta) = 0$ and $\mathbb{E}(\mathbf{x}) = 0$, $\hat{\theta} = \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{x}$.

Example

$$x[n] = A + w[n], \quad n = 0, 1, \ldots, N-1, \quad \text{where } A \sim \mathcal{U}[-A_0, A_0], w[n] \sim \mathcal{N}(0, \sigma^2)$$

MMSE estimator cannot be obtained in closed form.

LMMSE estimator:

$$E(\mathbf{x}) = \mathbf{0} \qquad \sigma_A^2 = E(A^2)$$

$$\begin{aligned} \mathbf{C}_{xx} &= E(\mathbf{x}\mathbf{x}^T) \\ &= E\left[(A\mathbf{1}+\mathbf{w})(A\mathbf{1}+\mathbf{w})^T\right] \\ &= E(A^2)\mathbf{1}\mathbf{1}^T + \sigma^2\mathbf{I} \end{aligned} \qquad \begin{aligned} \mathbf{C}_{\theta x} &= E(A\mathbf{x}^T) \\ &= E\left[A(A\mathbf{1}+\mathbf{w})^T\right] \\ &= E(A^2)\mathbf{1}^T \end{aligned}$$

$$\begin{aligned} \hat{A} &= \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{x} \\ &= \sigma_A^2\mathbf{1}^T(\sigma_A^2\mathbf{1}\mathbf{1}^T + \sigma^2\mathbf{I})^{-1}\mathbf{x} \end{aligned} \qquad \boxed{(\mathbf{A}+\mathbf{u}\mathbf{u}^T)^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{u}\mathbf{u}^T\mathbf{A}^{-1}}{1+\mathbf{u}^T\mathbf{A}^{-1}\mathbf{u}}}$$

$$\hat{A} = \frac{\sigma_A^2}{\sigma_A^2 + \frac{\sigma^2}{N}}\bar{x} = \frac{\frac{A_0^2}{3}}{\frac{A_0^2}{3} + \frac{\sigma^2}{N}}\bar{x} \qquad \sigma_A^2 = E(A^2) = (2A_0)^2/12 = A_0^2/3$$

- We don't care about the distribution of A.
- We only need to know its mean and variance to compute the LMMSE estimator.
- It is required that A and w are **uncorrelated** (as opposed to the independence assumption).
- But the LMMSE estimator is suboptimal.
- MMSE estimator:

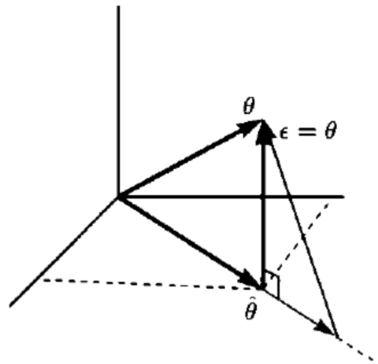
$$\begin{aligned} \hat{A} &= E(A|\mathbf{x}) \\ &= \int_{-\infty}^{\infty} A p(A|\mathbf{x})\, dA = \frac{\int_{-A_0}^{A_0} A \frac{1}{\sqrt{2\pi\frac{\sigma^2}{N}}} \exp\left[-\frac{1}{2\frac{\sigma^2}{N}}(A-\bar{x})^2\right] dA}{\int_{-A_0}^{A_0} \frac{1}{\sqrt{2\pi\frac{\sigma^2}{N}}} \exp\left[-\frac{1}{2\frac{\sigma^2}{N}}(A-\bar{x})^2\right] dA}. \end{aligned}$$

Geometrical Interpretations

- Vector space of random variables (assume **mean zero** vectors).
- Inner product between X and Y is E(XY).
- Random variables X and Y are orthogonal if E(XY) = 0 (i.e., uncorrelated).

LMMSE estimator $\hat{\theta} = \sum_{n=0}^{N-1} a_n x[n]$ minimizes

$$E[(\theta - \hat{\theta})^2] = E\left[\left(\theta - \sum_{n=0}^{N-1} a_n x[n]\right)^2\right] = \left\|\theta - \sum_{n=0}^{N-1} a_n x[n]\right\|^2.$$

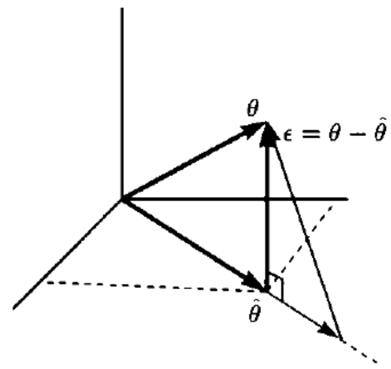


Orthogonality principle (a.k.a. projection theorem):

- $\hat{\theta}$ is in the subspace $\mathcal{S}$ spanned by $\{x[0], x[1], \ldots, x[N-1]\}$.
- Error vector $\epsilon = \theta - \hat{\theta}$ is minimized if ϵ is orthogonal to the subspace $\mathcal{S}$.
- $\epsilon \perp \mathcal{S}$
- $\epsilon \perp x[0], x[1], \ldots, x[N-1]$
- $\mathbb{E}[(\theta - \hat{\theta})x[n]] = 0$ for $n = 0, 1, \ldots, N-1$

Geometrical Interpretations

Orthogonality: $\mathbb{E}[(\theta - \hat{\theta})x[n]] = 0$ for $n = 0, 1, \ldots, N-1$



$$\mathbb{E}\left[\left(\theta - \sum_{m=0}^{N-1} a_m x[m]\right) x[n]\right] = 0 \qquad n = 0, 1, \ldots, N-1$$

$$\sum_{m=0}^{N-1} a_m \mathbb{E}(x[m]x[n]) = \mathbb{E}(\theta x[n]) \qquad n = 0, 1, \ldots, N-1$$

$$\begin{bmatrix} E(x^2[0]) & E(x[0]x[1]) & \ldots & E(x[0]x[N-1]) \\ E(x[1]x[0]) & E(x^2[1]) & \ldots & E(x[1]x[N-1]) \\ \vdots & \vdots & \ddots & \vdots \\ E(x[N-1]x[0]) & E(x[N-1]x[1]) & \ldots & E(x^2[N-1]) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{N-1} \end{bmatrix} = \begin{bmatrix} E(\theta x[0]) \\ E(\theta x[1]) \\ \vdots \\ E(\theta x[N-1]) \end{bmatrix}$$

$$\mathbf{C}_{xx}\mathbf{a} = \mathbf{C}_{x\theta}$$

$$\mathbf{a} = \mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta}$$

$$\begin{aligned} \hat{\theta} &= \mathbf{a}^T \mathbf{x} \\ &= \mathbf{C}_{x\theta}^T \mathbf{C}_{xx}^{-1} \mathbf{x} = \mathbf{C}_{\theta x} \mathbf{C}_{xx}^{-1} \mathbf{x} \end{aligned}$$

$$\begin{aligned} \mathrm{Bmse}(\hat{\theta}) &= E\left[\left(\theta - \sum_{n=0}^{N-1} a_n x[n]\right)\left(\theta - \sum_{m=0}^{N-1} a_m x[m]\right)\right] \\ &= E\left[\left(\theta - \sum_{n=0}^{N-1} a_n x[n]\right)\theta\right] - E\left[\left(\theta - \sum_{n=0}^{N-1} a_n x[n]\right)\sum_{m=0}^{N-1} a_m x[m]\right] \\ &= E(\theta^2) - \sum_{n=0}^{N-1} a_n E(x[n]\theta) - \cancel{\sum_{m=0}^{N-1} a_m E\left[\left(\theta - \sum_{n=0}^{N-1} a_n x[n]\right) x[m]\right]} \end{aligned}$$

Orthogonality principle

$$\begin{aligned} \mathrm{Bmse}(\hat{\theta}) &= C_{\theta\theta} - \mathbf{a}^T \mathbf{C}_{x\theta} \\ &= C_{\theta\theta} - \mathbf{C}_{x\theta}^T \mathbf{C}_{xx}^{-1} \mathbf{C}_{x\theta} \\ &= C_{\theta\theta} - \mathbf{C}_{\theta x} \mathbf{C}_{xx}^{-1} \mathbf{C}_{x\theta} \end{aligned}$$

Vector LMMSE Estimator

$$\hat{\theta}_i = \sum_{n=0}^{N-1} a_{in}x[n] + a_{iN} \quad \text{for } i = 1, 2, \ldots, p$$

minimizes the Bayes MSE for each i.

$$\hat{\theta}_i = E(\theta_i) + \mathbf{C}_{\theta_i x}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})) \qquad i = 1, 2, \ldots, p$$

$$\mathrm{Bmse}(\hat{\theta}_i) = C_{\theta_i\theta_i} - \mathbf{C}_{\theta_i x}\mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta_i} \qquad i = 1, 2, \ldots, p.$$

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= \begin{bmatrix} E(\theta_1) \\ E(\theta_2) \\ \vdots \\ E(\theta_p) \end{bmatrix} + \begin{bmatrix} \mathbf{C}_{\theta_1 x}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})) \\ \mathbf{C}_{\theta_2 x}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})) \\ \vdots \\ \mathbf{C}_{\theta_p x}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})) \end{bmatrix} \\ &= \begin{bmatrix} E(\theta_1) \\ E(\theta_2) \\ \vdots \\ E(\theta_p) \end{bmatrix} + \begin{bmatrix} \mathbf{C}_{\theta_1 x} \\ \mathbf{C}_{\theta_2 x} \\ \vdots \\ \mathbf{C}_{\theta_p x} \end{bmatrix} \mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})) \\ &= E(\boldsymbol{\theta}) + \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})) \end{aligned}$$

$$\begin{aligned} \mathbf{M}_{\hat{\theta}} &= E\left[(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^T\right] \\ &= \mathbf{C}_{\theta\theta} - \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta} \end{aligned}$$

$$\mathrm{Bmse}(\hat{\theta}_i) = [\mathbf{M}_{\hat{\theta}}]_{ii}$$

Properties of the LMMSE Estimator

1. Commutes over linear (affine) transformations

$$\boldsymbol{\alpha} = \mathbf{A}\boldsymbol{\theta} + \mathbf{b} \longrightarrow \hat{\boldsymbol{\alpha}} = \mathbf{A}\hat{\boldsymbol{\theta}} + \mathbf{b}$$

2. LMMSE estimator of a sum of unknown parameters is the sum of the individual estimators

$$\boldsymbol{\alpha} = \boldsymbol{\theta}_1 + \boldsymbol{\theta}_2 \longrightarrow \hat{\boldsymbol{\alpha}} = \hat{\boldsymbol{\theta}}_1 + \hat{\boldsymbol{\theta}}_2$$

$$\begin{aligned} \hat{\boldsymbol{\theta}}_1 &= E(\boldsymbol{\theta}_1) + \mathbf{C}_{\theta_1 x}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})) \\ \hat{\boldsymbol{\theta}}_2 &= E(\boldsymbol{\theta}_2) + \mathbf{C}_{\theta_2 x}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})). \end{aligned}$$

3. LMMSE estimator = MMSE estimator for linear Gaussian models, for which the MMSE estimator is linear in data

Theorem 12.1 (Bayesian Gauss-Markov Theorem) *If the data are described by the Bayesian linear model form*

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w} \tag{12.25}$$

where $\mathbf{x}$ *is an* $N \times 1$ *data vector,* $\mathbf{H}$ *is a known* $N \times p$ *observation matrix,* $\boldsymbol{\theta}$ *is a* $p \times 1$ *random vector of parameters whose realization is to be estimated and has mean* $E(\boldsymbol{\theta})$ *and covariance matrix* $\mathbf{C}_{\theta\theta}$*, and* $\mathbf{w}$ *is an* $N \times 1$ *random vector with zero mean and covariance matrix* $\mathbf{C}_w$ *and is uncorrelated with* $\boldsymbol{\theta}$ *(the joint PDF* $p(\mathbf{w}, \boldsymbol{\theta})$ *is otherwise arbitrary), then the LMMSE estimator of* $\boldsymbol{\theta}$ *is*

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= E(\boldsymbol{\theta}) + \mathbf{C}_{\theta\theta}\mathbf{H}^T(\mathbf{H}\mathbf{C}_{\theta\theta}\mathbf{H}^T + \mathbf{C}_w)^{-1}(\mathbf{x} - \mathbf{H}E(\boldsymbol{\theta})) && (12.26) \\ &= E(\boldsymbol{\theta}) + (\mathbf{C}_{\theta\theta}^{-1} + \mathbf{H}^T\mathbf{C}_w^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}_w^{-1}(\mathbf{x} - \mathbf{H}E(\boldsymbol{\theta})). && (12.27) \end{aligned}$$

The performance of the estimator is measured by the error $\boldsymbol{\epsilon} = \boldsymbol{\theta} - \hat{\boldsymbol{\theta}}$ *whose mean is zero and whose covariance matrix is*

$$\begin{aligned} \mathbf{C}_\epsilon &= E_{x,\theta}(\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T) \\ &= \mathbf{C}_{\theta\theta} - \mathbf{C}_{\theta\theta}\mathbf{H}^T(\mathbf{H}\mathbf{C}_{\theta\theta}\mathbf{H}^T + \mathbf{C}_w)^{-1}\mathbf{H}\mathbf{C}_{\theta\theta} && (12.28) \\ &= (\mathbf{C}_{\theta\theta}^{-1} + \mathbf{H}^T\mathbf{C}_w^{-1}\mathbf{H})^{-1}. && (12.29) \end{aligned}$$

The error covariance matrix is also the minimum MSE matrix $\mathbf{M}_{\hat{\theta}}$ *whose diagonal elements yield the minimum Bayesian MSE*

$$\begin{aligned} [\mathbf{M}_{\hat{\theta}}]_{ii} &= [\mathbf{C}_\epsilon]_{ii} \\ &= \mathrm{Bmse}(\hat{\theta}_i). && (12.30) \end{aligned}$$